9.2. ENZYME ACTION

Active Site:

It is a three dimensional area or region on the surface of the enzyme molecule where the substrate binds. It is also known as 'catalytic site' or 'substrate site'

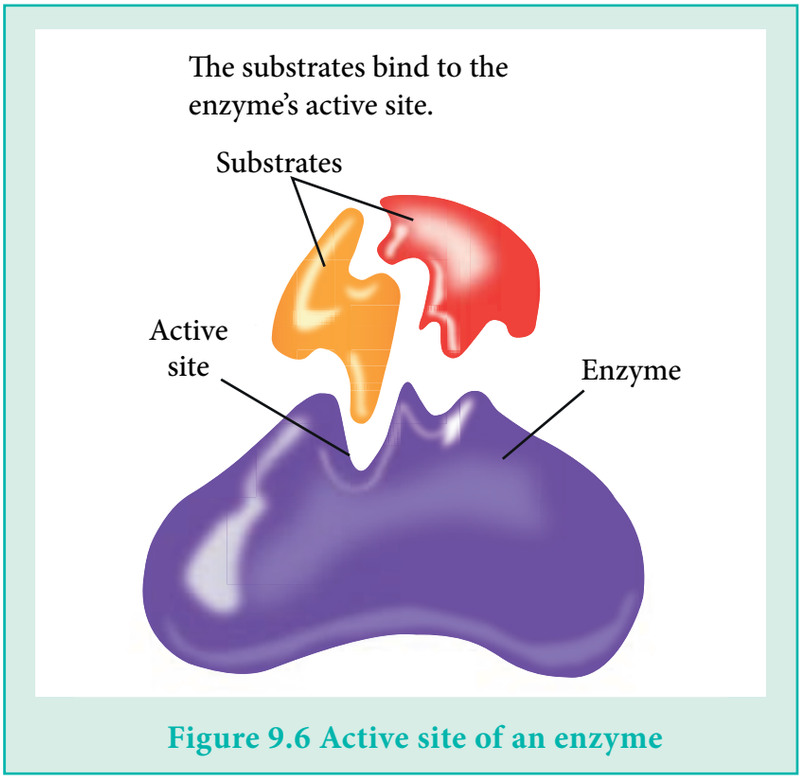


Figure 9.6 Active site of an enzyme

Although the enzymes differ widely in their properties, the active site present in their molecule possesses some common features. These are listed below :

The active site occupies a relatively small portion of the enzyme molecule.

The active site is neither a point nor a line or even a plane but is a 3- dimensional entity. It is made up of groups that come from different parts of the linear amino acid sequence.

Usually, the arrangement of atoms in the active site is well defined, resulting in a marked specificity of the enzymes. Although cases are known where the active site changes its configuration in order to bind a substance which is only slightly different in structure from its own substrate.

The active site binds the substrate molecule by relatively weak forces.

The active sites in the enzyme molecules are grooves or crevices from which water is largely excluded. It contains amino acids such as aspartic acid, glutamic acid, lysine, serine etc. The side chain groups like -COOH, $-NH_2$, $-CH_2OH$ etc., serve as catalytic groups in the active site. Besides, the crevice creates a micro-environment in which is essential for catalysis.

(i) Mode of enzyme action

In order to explain the mode of enzyme action, the following models have been proposed.

i) Fischer's lock and key model: According to this model the substrate binds with the enzyme at the active site just like a key fits into a lock. This leads to the formation of enzyme-substrate complex (ES complex). This complex decomposes to give the product and free enzyme.

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